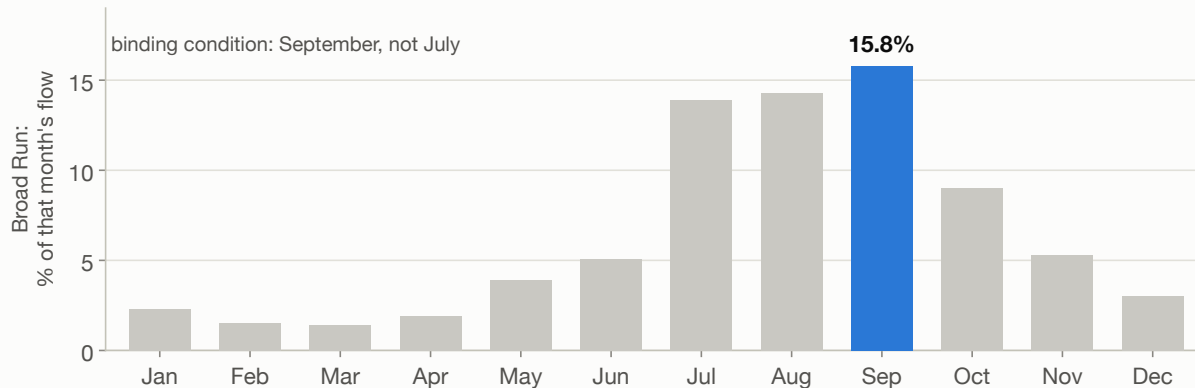
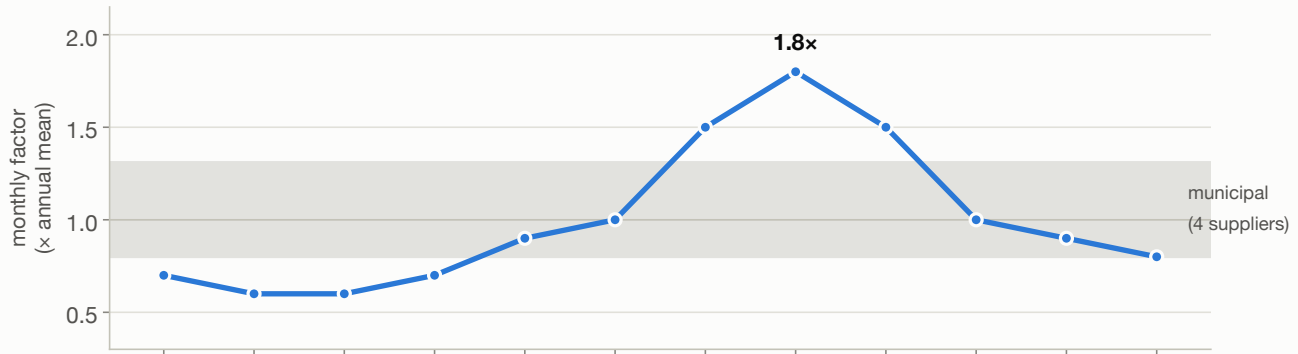


Demand peaks when the water isn't there

Data-center demand is ~6× peakier than the municipal demand the supply system was engineered around

- data centers (ICPRB observed, Loudoun + Prince William)
- municipal range (ICPRB Table 4-3, 11 yr of daily production)



Peak-DAY factors: municipal 1.5–1.9×; data centers 9.9× here and ~10× in ICPRB's independent construction — a 6.0× difference.

The monthly shape is ICPRB's OBSERVED Table A.3-2 series, which replaced a cooling-degree-day model that was ~70% too peaky in summer (July 3.04× against an observed 1.5×). Correcting it moved the binding condition from July to September and roughly halved it. The load is near-constant by design: JLARC records that Virginia data centers "do not currently participate in demand response programs" because "energy use is driven by computing activity."